

Synapse — Complete Real-Time Test Plan

End-to-end validation for the DU2 Hackathon demo stack. Focus: **live DIMSE intake**, **WebSocket events**, **Celery workers**, **STOW-RS routing**, **bulk migration**, and **UI refresh without page reload**.

Estimated duration: 90–120 minutes (full run) · **Smoke path:** 45 minutes (Sections 2, 3, 5.1, 6.1) · **Demo rehearsal:** 20 minutes (see [DEMO SCRIPT.md](#))

Phase coverage

Phase	Focus	Sections
0	Platform, health, seed data	2, 10
1	Admin config CRUD	11
2	DIMSE C-ECHO / C-STORE	5
3	Routing, STOW-RS, morphing	6
4	Bulk migration	7
5	Dashboard, reports, audit	8
6	Chatbot + PHI	9
—	RBAC, security	4

1. Test Environment

1.1 Prerequisites

Requirement	Detail
Docker Desktop	4.x+, Compose v2, 4 CPU / 8 GB RAM min
Ports free	3000, 8000, 8080, 8042, 8043, 11112, 11434, 4242
OS	Windows / macOS / Linux with Docker

1.2 Bootstrap

```
cd d:\PROJECTS\Synapse
cp .env.example .env
docker compose up --build -d
docker exec synapse-ollama ollama pull qwen2.5:7b-instruct
```

Wait until all containers are healthy:

```
docker compose ps
```

1.3 Service URLs

Service	URL
Frontend	http://localhost:3000
API / Swagger	http://localhost:8000/docs
Keycloak	http://localhost:8080
Orthanc On-Prem	http://localhost:8042
Orthanc Cloud	http://localhost:8043
WebSocket	ws://localhost:8000/ws/events
DIMSE	localhost:11112 (AE: SYNAPSE)

1.4 Test Users

Username	Password	Roles
admin	admin123	admin, operator, service_user, viewer
operator	operator123	operator, service_user, viewer
service	service123	service_user, viewer
viewer	viewer123	viewer

1.5 Obtain API Token (optional CLI tests)

```
curl -s -X POST "http://localhost:8080/realms/synapse/protocol/openid-connect/token" \
-H "Content-Type: application/x-www-form-urlencoded" \
-d "client_id=synapse-ui" \
-d "username=admin" \
-d "password=admin123" \
-d "grant_type=password" | jq -r .access_token
```

Export as `TOKEN` for curl commands below.

2. Pre-Flight Checklist (5 min)

#	Check	Command / Action	Pass Criteria
P1	All containers up	<code>docker compose ps</code>	10 services running
P2	API health	<code>curl http://localhost:8000/api/v1/health</code>	status: healthy OR degraded with only ollama model missing
P3	DIMSE listening	Health → <code>dimse_listener</code>	<code>SYNAPSE@11112</code>

P4	Celery routing	<code>docker logs synapse-celery-routing --tail 5</code>	Worker ready
P5	Celery migration	<code>docker logs synapse-celery-migration --tail 5</code>	Worker ready
P6	Seed data	UI → Nodes	Orthanc On-Prem + Orthanc Cloud present
P7	Routing rule	UI → Routing Rules	"Route CT to Cloud PACS" active
P8	Morphing rule	UI → Tag Morphing	"CT Institution Rename" active
P9	Ollama model	<code>docker exec synapse-ollama ollama list</code>	qwen2.5:7b-instruct listed
P10	UI login	http://localhost:3000	Synapse theme, dashboard loads

3. Automated Unit Tests (5 min)

```
docker build -t synapse-backend ./backend
docker run --rm synapse-backend python -m pytest tests/ -v
```

Pass: All tests green (22+).

```
cd frontend && npm run build
```

Pass: Build succeeds with no errors.

4. Authentication & RBAC (10 min)

4.1 Keycloak Login Theme

#	Step	Pass Criteria
A1	Logout, visit http://localhost:3000	Redirect to Keycloak
A2	Observe login page	Synapse branding, dark theme
A3	Login as admin / admin123	Redirect to dashboard

4.2 Role-Based Navigation

Test each user; verify sidebar items match role:

Role	Must see	Must NOT see
viewer	Dashboard, Chatbot	Nodes, Settings, Migration Jobs, Reports

service	Dashboard, Routing Monitor, Audit Logs, Reports, Chatbot	Nodes, Settings
operator	+ Migration Jobs, System Health	Nodes, Settings
admin	All pages	—

4.3 Login Audit

#	Step	Pass Criteria
A4	Login as any user	Audit Logs → USER_LOGIN event within 30s

5. Real-Time DIMSE Intake (15 min)

Reference: [DIMSE_TESTING.md](#)

5.1 C-ECHO + C-STORE (Primary Path)

Terminal 1 — open Routing Monitor in browser (WebSocket pill = **live**).

Terminal 2 — send study:

```
docker exec synapse-backend python scripts/test_dimse_e2e.py --host localhost --port 11112 -  
-instances 3
```

#	Real-time observation	Pass Criteria	Timing
D1	Routing Monitor → Live Events	study_received appears	< 5s
D2	DIMSE metrics cards	Studies Assembled +1	< 10s
D3	Live Events	routing_completed with status	< 30s
D4	Routing Transactions table	New row, status success	< 30s
D5	Dashboard metrics	Studies Routed increments	< 15s (auto-refresh)
D6	Audit Logs	STUDY_RECEPTION, ROUTING_RULE_MATCH, TAG_MORPHING_APPLIED	< 60s

```
curl -s -H "Authorization: Bearer $TOKEN" http://localhost:8000/api/v1/dimse/status | jq
```

Pass: listening: true , statistics.studies_assembled increased.

5.2 WebSocket Event Stream (API-level)

Use browser DevTools → Network → WS on ws://localhost:8000/ws/events , or:

```
// Browser console on localhost:3000
const ws = new WebSocket("ws://localhost:8000/ws/events");
ws.onmessage = (e) => console.log(JSON.parse(e.data));
```

Send C-STORE again. **Pass:** JSON messages with `study_received` and `routing_completed`.

5.3 Promiscuous Mode / AE Rejection

#	Step	Pass Criteria
D7	Settings → promiscuous mode OFF	Saved
D8	<code>docker exec synapse-backend python scripts/test_dimse_e2e.py --calling-ae UNKNOWN_AE</code>	Association rejected
D9	DIMSE status → recent events	association_rejected
D10	Enable promiscuous mode	Re-test succeeds

5.4 No-Match Routing

Generate MR study (no rule):

```
docker exec synapse-backend python scripts/generate_test_dicom.py --output
/data/temp_dicom/mr_test --modality MR --instances 1
docker exec synapse-backend python -c "
from pathlib import Path
from pynetdicom import AE
from pydicom import dcmread
files = list(Path('/data/temp_dicom/mr_test').glob('*.dcm'))
ae = AE(ae_title='STORESCU')
ds = dcmread(files[0])
ae.add_requested_context(ds.SOPClassUID)
assoc = ae.associate('localhost', 11112, ae_title='SYNAPSE')
assoc.send_c_store(ds)
assoc.release()
print('MR sent')
"
```

Pass: Transaction `overall_status` = `no_match` ; no STOW-RS to cloud.

6. Real-Time Routing & STOW-RS (15 min)

Reference: [ROUTING_TESTING.md](#)

6.1 End-to-End Protocol Conversion

Modality --C-STORE (DIMSE)--> Synapse --STOW-RS (DICOMweb)--> Cloud Orthanc

#	Verification	How	Pass Criteria
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R1	Cloud PACS received study	http://localhost:8043 → Studies	CT study visible
R2	Tag morphing applied	Cloud Orthanc metadata / QIDO	InstitutionName = Cloud Demo Hospital
R3	No DIMSE to destination	Cloud orthanc.json has no DIMSE port exposed	Only DICOMweb path used

```
curl -s "http://localhost:8043/dicom-web/studies?Modality=CT" -H "Accept: application/dicom+json" | jq length
```

6.2 Per-Destination Retry (Failure Recovery)

#	Step	Pass Criteria
R4	docker compose stop orthanc-cloud	Cloud down
R5	Send new CT study via DIMSE	Destination status failed in UI
R6	docker compose start orthanc-cloud	Cloud back
R7	Click Retry on failed destination	Status → success
R8	Audit Logs	RETRY_ATTEMPT event

API alternative:

```
curl -X POST -H "Authorization: Bearer $TOKEN" \
  "http://localhost:8000/api/v1/routing-transactions/destinations/{DEST_ID}/retry"
```

6.3 Multi-Destination (Optional)

#	Step	Pass Criteria
R9	Add second cloud destination node (admin)	Node created
R10	Edit routing rule → two destinations	Rule saved
R11	Send CT study	Two destination rows, independent statuses

7. Bulk Migration — Real-Time Progress (20 min)

Reference: [MIGRATION TESTING.md](#)

7.1 Prepare Source Data

Upload studies to on-prem Orthanc (if empty):

```
# Generate + C-STORE to on-prem via DIMSE, OR use Orthanc UI at :8042 to upload
docker exec synapse-backend python scripts/test_dimse_e2e.py --instances 2
```

```
# Studies land in Synapse temp; for migration source use Orthanc on-prem DICOMweb
```

Ensure on-prem Orthanc has at least one CT study (upload via <http://localhost:8042>).

7.2 Create & Start Migration Job

#	Step (UI as operator)	Pass Criteria
M1	Migration Jobs → New Job	Form opens
M2	Source: Orthanc On-Prem, Dest: Orthanc Cloud, filter Modality=CT	Job created not_started
M3	Click Start	Status → in_progress
M4	Open job Details	Study records appear (QIDO discovery)
M5	Watch progress counters	completed_studies increments live (5s refresh)
M6	Celery logs	<code>docker logs -f synapse-celery-migration</code> shows <code>migrate_study</code>
M7	Job completes	Status completed OR partial
M8	Cloud Orthanc	Migrated studies visible

7.3 Cancel & Resume

#	Step	Pass Criteria
M9	Start new job, click Cancel mid-run	Status cancelled
M10	Job with failures or partial status, click Resume (re-calls POST /start)	Job re-enters in_progress; failed/pending studies re-enqueued

Note: Pause mid-job is not implemented. Use **Cancel** to stop, or **Resume** on `failed` / `partial` jobs to restart processing.

7.4 Per-Study Retry

#	Step	Pass Criteria
M11	Force failure (stop cloud during job)	Study record failed
M12	Restart cloud, click Retry on study	Status → success

8. Dashboard & Reporting — Live Metrics (10 min)

Reference: [REPORTING.md](#)

#	Page	Real-time test	Pass Criteria
DR1	Dashboard	Send C-STORE while page open	Metrics + activity feed update within 10–15s

DR2	Dashboard	Volume chart (7 days)	Bar for today increments
DR3	Reports	Change period 7→30 days	Summary recalculates
DR4	Audit Logs	Filter by ROUTING_RULE_MATCH	Matching events only
DR5	Audit Logs	Export CSV	File downloads
DR6	Reports	Export Audit CSV	File downloads (operator/admin)

```
curl -s -H "Authorization: Bearer $TOKEN" http://localhost:8000/api/v1/dashboard/metrics |  
jq  
curl -s -H "Authorization: Bearer $TOKEN" "http://localhost:8000/api/v1/reports/summary?  
days=7" | jq
```

9. Chatbot (10 min)

Reference: [CHATBOT.md](#)

#	Step	Pass Criteria
C1	Chatbot page → status pill	"Ollama ready" (or fallback if model missing)
C2	Ask: "What is the migration status?"	Coherent answer using live counts
C3	Ask: "How many studies failed routing today?"	References routing data
C4	Ask: "Is the DIMSE listener online?"	Correct listener state
C5	Include Study UID in question	Study-specific lookup in answer
C6	Audit Logs	CHATBOT_QUERY events
C7	Login as viewer	PHI redacted (masked UIDs); banner shown
C8	Login as service	Full operational detail in answers
C9	Stop Ollama: docker compose stop ollama	Fallback response + "Fallback" badge
C10	Restart Ollama	LLM responses resume

10. System Health & Workers (5 min)

#	Check	Pass Criteria
H1	UI → System Health	All components healthy (ollama may warn if model not pulled)
H2	Stop postgres → refresh health	postgresql unhealthy, overall degraded
H3	Restart postgres	Recovers to healthy
H4	docker logs synapse-backend --tail 20	No repeated tracebacks

H5	<code>docker logs synapse-celery-routing --tail 20</code>	route_study tasks complete
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11. Admin Configuration CRUD (10 min)

#	Area	Test	Pass Criteria
AD1	Nodes	Create / edit / delete test node	Audit CONFIG_CHANGE
AD2	Routing Rules	Create rule, preview	Preview returns match result
AD3	Tag Morphing	Preview morphing	Shows before/after
AD4	Settings	Change DIMSE AE title (careful)	Requires stack restart to fully apply listener

12. Negative & Edge Cases

#	Scenario	Expected behavior
N1	API without token	401 Unauthorized
N2	Viewer calls POST /nodes	403 Forbidden
N3	Operator calls GET /nodes	200 (read for migration UI)
N4	Invalid Study UID in chatbot	"No matching records" or honest "don't know"
N5	Duplicate C-STORE same Study UID	New routing transaction (idempotent at PACS level depends on Orthanc)
N6	Redis down	WebSocket events stop; routing may fail — health degraded

13. Real-Time Event Matrix

Events that should appear **without manual page refresh** (Routing Monitor open):

Trigger	WebSocket event_type	UI location	API
C-STORE complete	study_received	Live Events	/routing-transactions
STOW-RS done	routing_completed	Live Events + txn status	/routing-transactions
Migration study done	migration_study_completed	Dashboard activity (on refresh)	/migration-jobs/{id}/studies
Migration job done	migration_job_completed	Migration Jobs list	/migration-jobs

14. Demo Rehearsal Timeline (20 min)

Time	Activity
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0:00	Architecture slide: DIMSE → DICOMweb only to cloud
0:02	System Health — all green
0:03	Show Nodes, Routing Rules, Tag Morphing
0:06	Open Routing Monitor → send C-STORE → live events
0:11	Cloud Orthanc — verify study + morphed InstitutionName
0:13	Audit Logs — routing + morphing events
0:15	Migration Job — start, show progress
0:18	Chatbot — 2 questions; viewer PHI demo
0:20	RBAC — viewer vs operator nav

15. Pass / Fail Summary

Category	Minimum to pass demo
Infrastructure	8/10 pre-flight checks
DIMSE + Routing	D1–D6, R1–R3 all pass
Real-time UI	WebSocket live + transaction success
Migration	M1–M8 pass
Security	RBAC + viewer PHI redaction
Chatbot	C2–C4 pass (C9 fallback acceptable)
Audit	All major actions logged

16. Troubleshooting Quick Reference

Symptom	Likely cause	Fix
WebSocket offline	Backend/redis down	<code>docker compose restart backend redis</code>
Routing stuck pending	Celery routing worker	<code>docker compose restart celery-routing</code>
<code>no_match</code> for CT	Seed data missing	<code>docker compose down -v</code> and re-up, or create rules manually
Migration 0 studies	On-prem Orthanc empty	Upload studies to :8042
Chatbot fallback only	Ollama model not pulled	<code>docker exec synapse-ollama ollama pull qwen2.5:7b-instruct</code>

STOW-RS failed	Cloud Orthanc down	<code>docker compose restart orthanc-cloud</code>
Keycloak theme old	Cached realm	Set login theme to <code>synapse</code> in admin console

17. Related Docs

- [SETUP.md](#) — installation
- [DIMSE TESTING.md](#) — DIMSE scripts
- [ROUTING TESTING.md](#) — STOW-RS validation
- [MIGRATION TESTING.md](#) — bulk migration
- [REPORTING.md](#) — dashboard & audit export
- [CHATBOT.md](#) — Ollama chatbot
- [DEMO SCRIPT.md](#) — presentation flow

18. Printable Master Checklist

Copy this section for hackathon sign-off. Check each box during a live run.

Platform

- ☐ `docker compose up -d` — all core containers running
- ☐ `curl http://localhost:8000/api/v1/health` — dimse, postgres, redis, orthanc x2, keycloak healthy
- ☐ Ollama model pulled (`qwen2.5:7b-instruct`) or chatbot fallback acceptable
- ☐ Keycloak login + Synapse theme works
- ☐ Seed: Orthanc On-Prem, Orthanc Cloud, CT routing rule, morphing rule present
- ☐ `docker exec synapse-backend python -m pytest tests/ -v` — all pass

DIMSE (Phase 2)

- ☐ C-ECHO OK (`test_dimse_e2e.py --skip-store`)
- ☐ C-STORE → `study_assembled` in Routing Monitor DIMSE events
- ☐ Routing transaction row created
- ☐ WebSocket `study_received` without manual refresh
- ☐ Promiscuous OFF rejects unknown AE; ON accepts

Routing (Phase 3)

- ☐ CT study → cloud Orthanc via STOW-RS (not DIMSE to destination)
- ☐ Transaction status `success`
- ☐ `InstitutionName` = **Cloud Demo Hospital** in cloud PACS
- ☐ Audit: `ROUTING_RULE_MATCH` , `TAG_MORPHING_APPLIED`
- ☐ MR study → `no_match` (optional)
- ☐ Failed destination **Retry** succeeds after cloud Orthanc restart

Migration (Phase 4)

- ☐ On-prem Orthanc has source studies
- ☐ Job create + **Start** from Migration Jobs UI

- ☐ Progress counters update live
- ☐ Studies visible in cloud Orthanc after completion
- ☐ **Cancel** mid-run works
- ☐ **Resume** on failed/partial job works
- ☐ Per-study **Retry** works

Reporting (Phase 5)

- ☐ Dashboard metrics/charts reflect test activity
- ☐ Activity feed updates after routing
- ☐ Reports summary + audit CSV export
- ☐ Audit Logs: filters, chart, CSV, `USER_LOGIN`

Chatbot (Phase 6)

- ☐ Chatbot status shows Ollama ready (or fallback badge)
- ☐ Migration / routing / DIMSE questions answered from live data
- ☐ `CHATBOT_QUERY` in audit logs
- ☐ Viewer: PHI redacted; service user: full detail
- ☐ Ollama stopped → fallback response (optional)

Security & real-time

- ☐ RBAC: viewer / service / operator / admin nav matches matrix (Section 4.2)
- ☐ Viewer blocked from admin APIs (403)
- ☐ Logout clears session
- ☐ Routing completes while migration job runs (separate Celery queues)

Quick smoke commands

```
docker compose up -d
docker exec synapse-ollama ollama pull qwen2.5:7b-instruct
docker exec synapse-backend python -m pytest tests/ -v
docker exec synapse-backend python scripts/test_dimse_e2e.py --host localhost --port 11112 -
-instances 3
```